Aditya Gambhir

Data Scientist | Riverside, CA, USA | +1 (951) 544-8605 | gambhir.aditya19@gmail.com | aditya-gambhir-portfolio.vercel.app/ | linkedin.com/in/aditya-gambhir | github.com/Aditya-gam

Education

University of California, Riverside Mar 2025

Master of Science | Computational Data Science

Relevant Coursework: Big Data Management, Advanced Operating Systems, Database Systems, Artificial Intelligence, Machine Learning, Spatial Computing, Data Analytics

Vishwakarma Institute of Information Technology May 2023

Bachelor of Technology - BTech | Computer Engineering

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Compiler Construction, Computer Architecture, Theory of Computation,

Skills

Languages: Java · C · C++ · Javascript · TypeScript · SQL · Python · Bash · R

Libraries/Frameworks: Machine Learning · NumPy · Pandas · Matplotlib · Seaborn · SpaCy · OpenCV · Scikit-Learn · Tensorflow · Keras · Pytorch

Databases: PostgreSQL · MySQL · MongoDB · SQLite

Cloud/DevOps: AWS · GCP · GitLab CI/CD · Azure · Docker · Kubernetes · Jenkins CI/CD · Continuous Integration · Continuous Delivery

Data Analytics/Big Data: Hadoop · Spark · Kafka · BigQuery · Snowflake · Data Mining · A/B Testing · Statistical Data Analysis

Tools: VS code · IntelliJ IDEA · Postman · Git · Jupyter Notebook · Excel · Tableau · Power BI

Web Development: React.js · Node.js · REST · Next.js · HTML · CSS

Experience

Tech Mahindra **Pune, Maharashtra**

Software Development Intern Jun 2022 - May 2023

- Reduced API integration time by 15% through collaborative build of a logistics simulation server, leading to 500+ developers completing integration tests 2 days faster.
- Implemented statistical anomaly detection on backend services, leveraging Jenkins CI/CD pipelines to flag 5 critical integration errors and improving overall system reliability by 20%.
- Implemented OAuth 2.0 and JWT authentication, enhancing data pipeline security across 10+ microservices and resolving five critical vulnerability alerts; reduced potential data breaches by an estimated 22%.

Centre for Development of Advanced Computing **Remote**

Research and Development Intern Jan 2022 - Jun 2022

- Engineered a robust backend service leveraging FastAPI, PostgreSQL, and React.js, enabling real-time analysis of sensor data streams with sub-500ms latency and enhanced data visualization.
- Established an audit trail capturing all data access and modification events, facilitating forensic and root cause analysis of data anomalies by 20%.
- Collaborated with senior engineers to design error handling strategies, guaranteeing service availability of 98% during peak loads and preventing data loss.

Projects

Sensor Fusion for Self-Driving Vehicles Oct 2024

C++, Point Cloud Library, Eigen, Kalman Filter

- Applied Unscented Kalman Filter for time-series sensor fusion and statistical analysis of tracking errors, achieving vehicle-tracking RMSE < 0.5 m.
- Processed point-cloud and radar feeds to enhance autonomous-driving safety and situational awareness.
- Modified sensor fusion algorithms within the highway simulation, producing a 20% decrease in system response delay.

Transformer Model for Crowd Localization Jun 2024

PyTorch, Transformers, Computer Vision

- Built transformer model in PyTorch, improving accuracy by 12% and reducing ambiguous detections by 20%.
- Optimized predictions with Hungarian matcher; validated robustness across five major datasets.
- Delivered a scalable crowd-localization pipeline for real-world deployment and performed statistical analysis of precision/recall trade-offs.

Chatbot Using Transformer Models Dec 2023

NLP, Deep Learning, Transformer, LSTM

- Achieved BLEU 25.67 with the Transformer model, 35% higher than the LSTM baseline.
- Optimized cloud compute infrastructure using cost-saving measures, reducing monthly expenditure by 15% while preprocessing large text datasets that enabled the creation of a domain-specific vocabulary of 8,245 words.
- Collaborated with a PhD researcher, providing insightful data analysis that identified and refined model training data with 5000+ samples, improving efficiency by 15%.